

X2 T1 Binary numbers a) What do all binary representations of numbers divisible by four have in common?

the last two bits = 00

0 divisible by four (0)

1 not divisible by four (1)

10 not divisible by four (2)

11 not divisible by four ($2+1=3$)

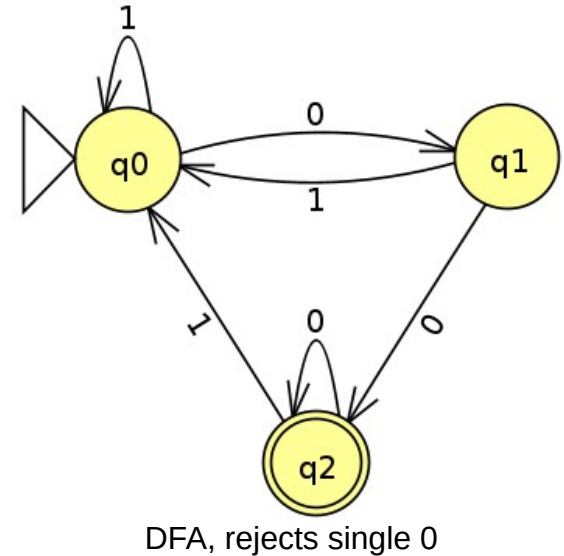
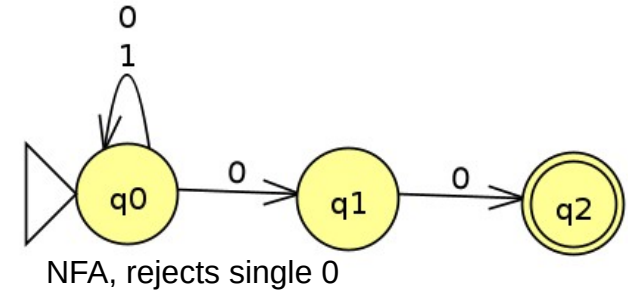
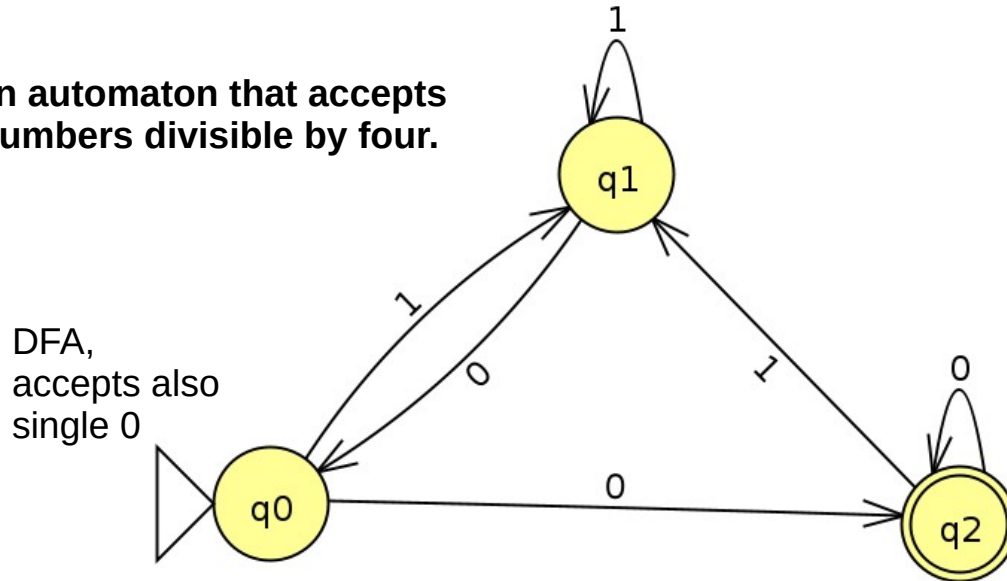
100 divisible by four (4)

1000 divisible by four (8)

1001 not divisible by four ($8+1=9$)

1100 divisible by four ($8+4=12$)

b) Create an automaton that accepts all binary numbers divisible by four.



X2 T2. Minimize the automaton according to the method presented in the lecture. Check the result with JFLAP.

There is no transition from any state to state q_1 , so it is an unreachable state and can be removed as unnecessary.

$$\delta(q_0, 0) = q_3, \delta(q_0, 1) = q_4$$
$$\delta(q_3, 0) = q_3, \delta(q_3, 1) = q_2$$

$$\delta(q_2, 0) = q_6, \delta(q_2, 1) = q_5$$
$$\delta(q_4, 0) = q_5, \delta(q_4, 1) = q_5$$

$$\delta(q_6, 0) = q_5, \delta(q_6, 1) = q_5$$
$$\delta(q_5, 0) = q_6, \delta(q_5, 1) = q_6$$

$$P_0 = \{\{q_0, q_2, q_3, q_4\}, \{q_5, q_6\}\}$$

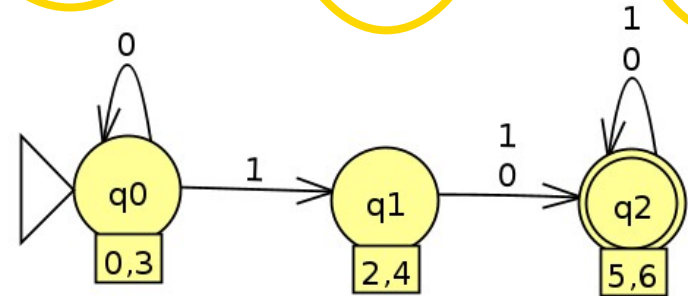
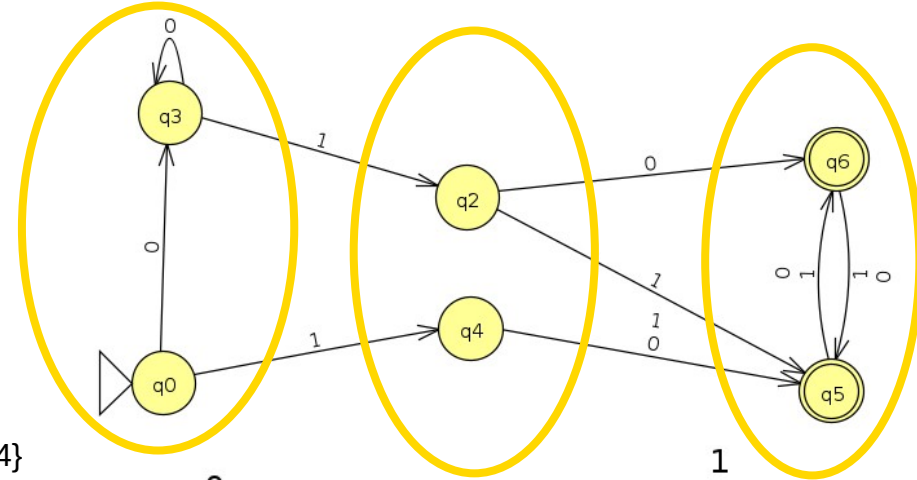
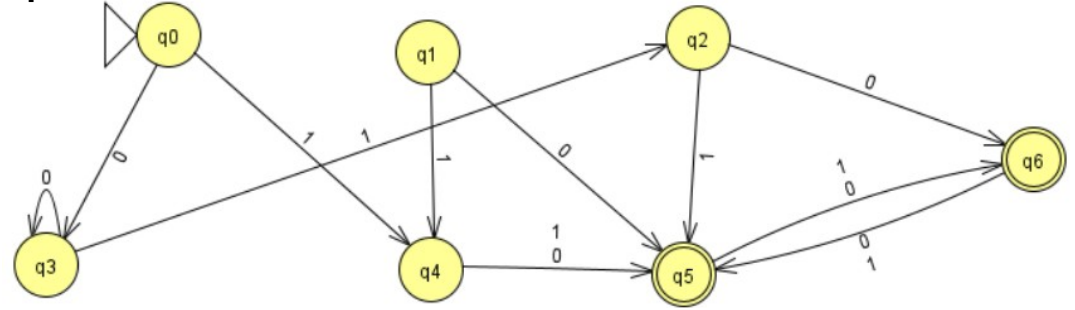
q_5 and q_6 stay together $\leftarrow$ for both inputs there's transition to the same set

q_2 and q_4 belong together
 $\leftarrow$ from both states the transitions lead to the set $\{q_5, q_6\}$ for both inputs

q_0 and q_3 belong together
 $\leftarrow$ with input 0 they stay in their own group, input 1 leads to the group $\{q_2, q_4\}$

$$P_1 = \{\{q_0, q_3\}, \{q_2, q_4\}, \{q_5, q_6\}\}$$

$\rightarrow$ the minimized automaton has three states.

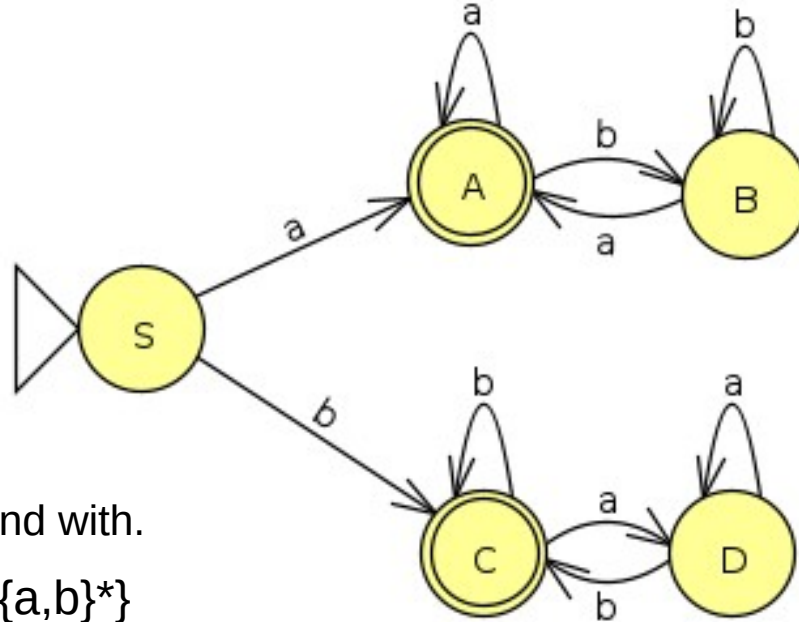


X2 T3. Create an automaton corresponding to the grammar below. What could be the language accepted by the automaton? The start symbol of the grammar is S.

$S \rightarrow aA \mid bC$
 $A \rightarrow aA \mid bB \mid \lambda$
 $B \rightarrow bB \mid aA$
 $C \rightarrow aD \mid bC \mid \lambda$
 $D \rightarrow aD \mid bC$

Strings that begin with the same character they end with.

$L = \{w \in \{a,b\}^* \mid (w = aza) \vee (w = bzb), z \in \{a,b\}^*\}$

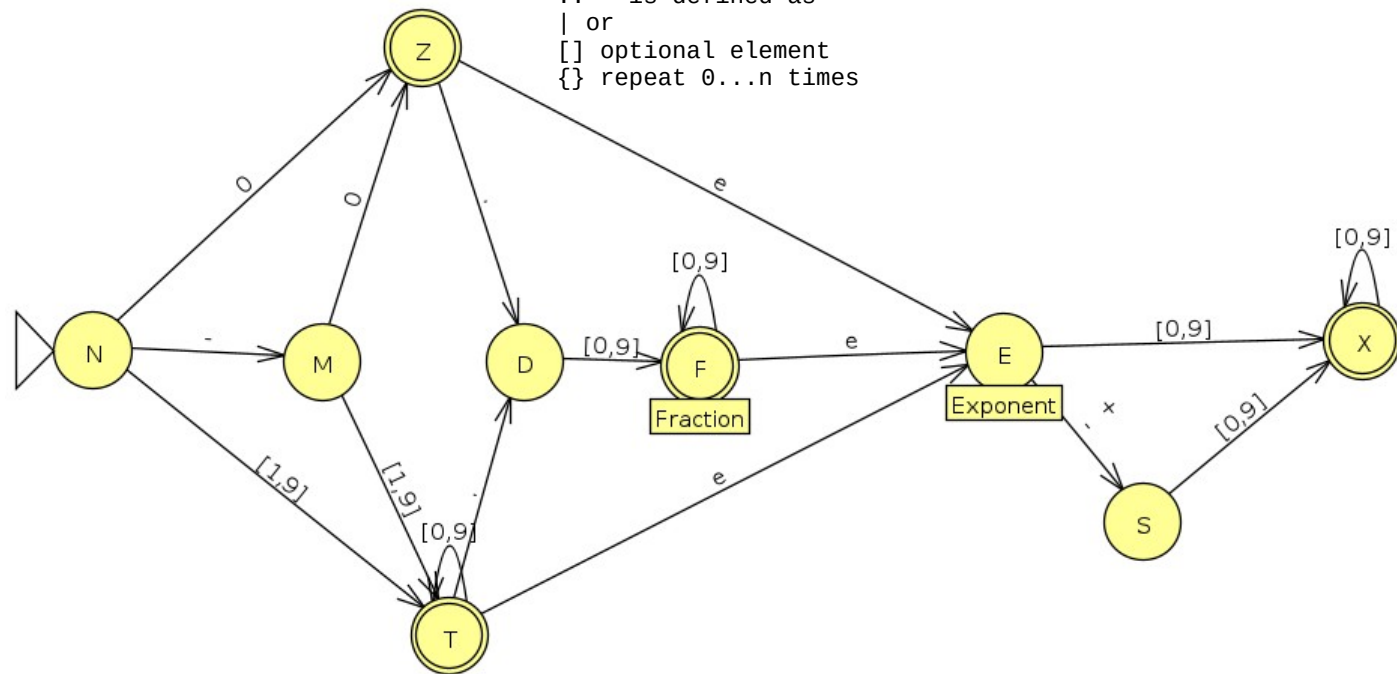
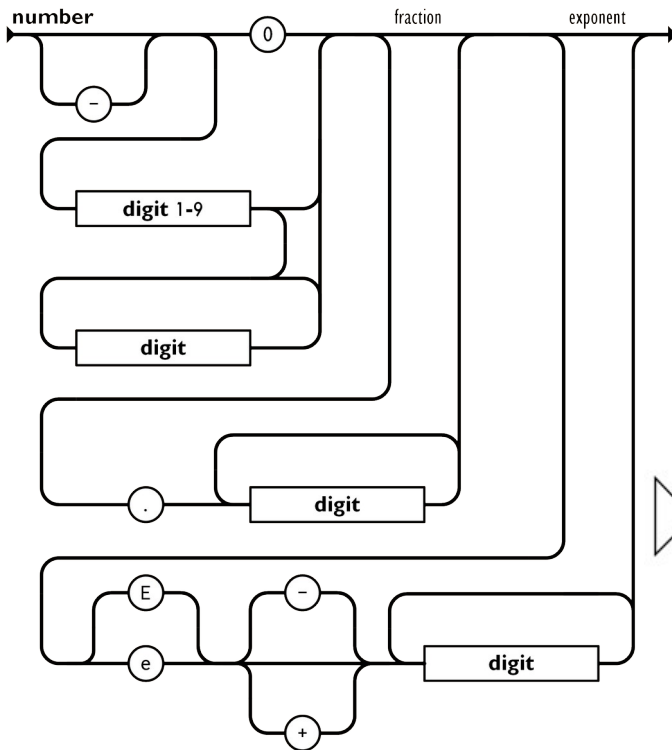


X2 T4. Familiarize yourself with the JSON syntax for number representation. <http://www.json.org/json-en.html>
On the page, the syntax of number is presented as a railroad diagram. Create a DFA corresponding to the railroad diagram, i.e., an automaton that accepts JSON numbers.

EBNF-format

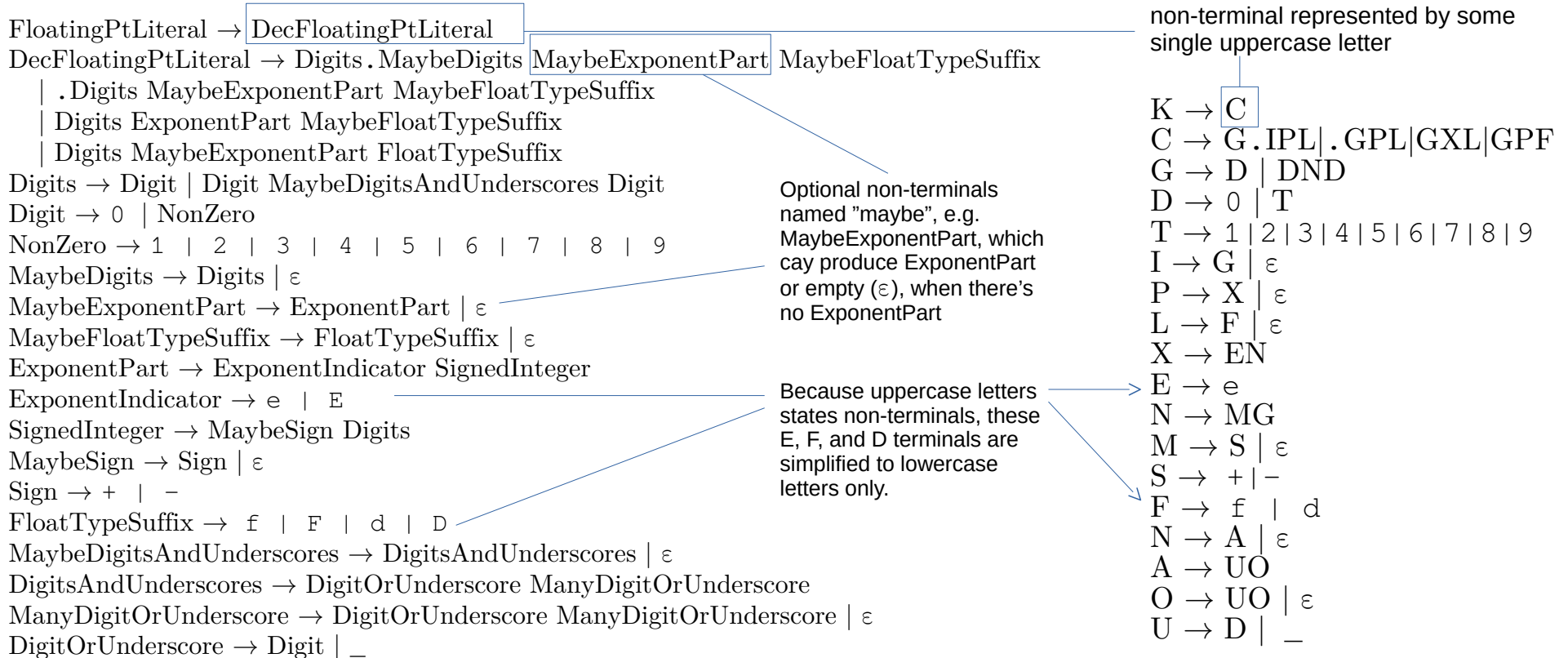
```
number ::= [ "-" ] int [ frac ] [ exp ]
int     ::= "0" | digit1-9 { digit }
digit1-9 ::= 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
frac    ::= "." digit { digit }
exp     ::= ("e" | "E") [ "+" | "-" ] digit { digit }
```

::= "is defined as"
 | or
 [] optional element
 {} repeat 0...n times



X2 T5. Familiarize yourself with the grammar notation used in the Java documentation and the representation of floating-point literals (<https://docs.oracle.com/javase/specs/jls/se17/html/jls-3.html#jls-3.10.2> , Floating-Point Literals). A floating-point literal is a number represented in decimal form and may contain an integer part, a decimal part, and an exponent (e.g., 123.45e-6).

Based on the grammar, create a simplified grammar for DecimalFloatingPointLiteral using the notation used in the lecture: Each nonterminal is represented by a single uppercase letter and each input symbol by a lowercase letter. The right side of the arrow shows what the nonterminal on the left side can produce (alternatives are separated by a vertical bar |). Repetition is produced recursively using the empty symbol (ϵ)(for example, $Y \rightarrow 1Y \mid \epsilon$ repeats the symbol 1 zero to n times)."



X2 T6. The string x accepted by the automaton consists of the characters abc , i.e., $\Sigma=\{a,b,c\}$. Among the last three characters of the string, there must be at least one a . Implement a nondeterministic automaton that accepts the language. Epsilon transitions may be used in a nondeterministic automaton.

